

Econ 381: Introduction to Econometrics

Handout 2: Example on Class Size Effects

Muyang Ren

Department of Economics
University of Tennessee, Knoxville

A first look at an econometric study

In this handout, we conduct a simple empirical analysis of the relationship between class size and students' test-score performance.

The purpose is to preview the questions that guide econometric analysis:

- ▶ What effect are we interested in?
- ▶ What variables can we observe in the data?
- ▶ How can an economic equation represent the relationship?
- ▶ What part of the relationship is unknown?
- ▶ How can we use data to estimate that unknown quantity?

We will study the statistical tools behind each step later in the course.

The superintendent faces a decision

A school-district superintendent is considering hiring more teachers.

Hiring more teachers would reduce the number of students per teacher and may produce several benefits:

- ▶ teachers could give students more individual attention;
- ▶ classrooms may have fewer disruptions; and
- ▶ students may receive more frequent feedback.

But additional teachers are costly. The superintendent must compare the expected improvement in student learning with the cost of reducing class size.

This is an economic cost–benefit question.

The research question

To make the decision, the superintendent wants to know:

How does reducing class size change student achievement?

We measure class size using the **student–teacher ratio**:

$$STR = \frac{\text{total student enrollment}}{\text{number of teachers}}.$$

- ▶ A lower *STR* means fewer students per teacher.
- ▶ A higher *STR* means more students per teacher.

Student achievement is measured using average test scores.

What is one observation in the data?

The California dataset contains **420 school districts**, randomly sampled from approximately 1,000 school districts.

Each observation represents one district:

$$i = 1, 2, \dots, 420.$$

These 420 school districts form one **sample**.

For every district i , researchers observe several district-level measurements.

For example, one observation records:

- ▶ the district's average test score;
- ▶ its student–teacher ratio;
- ▶ the percentage of students who are English learners;
- ▶ the percentage eligible for subsidized lunch;
- ▶ average district income; and
- ▶ district enrollment.

A few observations from the California data

i	School district	$TestScore_i$	STR_i	English learners (%)
1	Sunol Glen Unified	690.80	17.89	0.00%
2	Manzanita Elementary	661.20	21.52	4.58%
3	Thermalito Union Elementary	643.60	18.70	30.00%
4	Golden Feather Union Elementary	647.70	17.36	0.00%
5	Palermo Union Elementary	640.85	18.67	13.86%

Each row is one district. Notice that districts with similar STR can have quite different test scores and student characteristics.

Source: first five observations in `caschool.xlsx`, the 420-district dataset used in this example.

The two variables at the center of our question

Variable	Role	Measurement
$TestScore_i$	Outcome	District i 's average of reading and mathematics test scores
STR_i	Explanatory variable	District i 's total enrollment divided by its number of teachers

Our data therefore contain 420 observed pairs:

$$(TestScore_i, STR_i), \quad i = 1, \dots, 420.$$

The additional variables describe other differences across districts. They will help us think about why test scores can differ even when districts have similar class sizes.

Step 1: Describe the superintendent's policy change

Suppose the superintendent wants to reduce the student–teacher ratio.

Let

$$\Delta STR = STR_{\text{after}} - STR_{\text{before}}$$

denote the targeted change in the student–teacher ratio.

If the superintendent reduces STR by one student per teacher, then

$$\Delta STR = -1.$$

The negative sign is important: the policy moves the student–teacher ratio downward.

Step 2: Describe the outcome we want to predict

Let

$$\Delta TestScore = TestScore_{\text{after}} - TestScore_{\text{before}}$$

denote the resulting change in test scores.

If smaller classes improve learning, we would expect

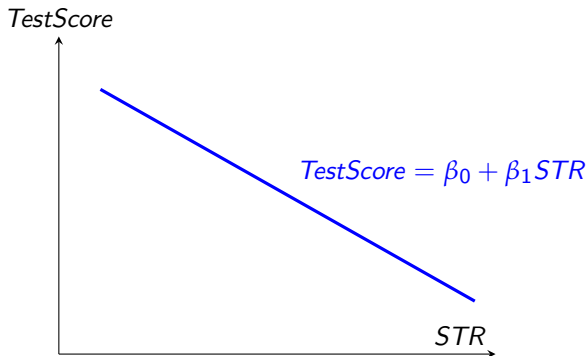
negative ΔSTR leads to positive $\Delta TestScore$

The superintendent's practical question is now:

If $\Delta STR = -1$, what is $\Delta TestScore$?

Step 3: Define the effect per unit change

Assume for now that test scores have a linear relationship with STR .



This line represents the systematic relationship we would like to identify. Its negative slope means that higher STR is associated with lower test scores.

Step 3 continued: Summarize the target effect by the slope

The target effect can be summarized by β_1 , the ratio of the change in test scores to the change in *STR*:

$$\beta_1 = \frac{\Delta TestScore}{\Delta STR}$$

Equivalently,

$$\Delta TestScore = \beta_1 \Delta STR.$$

- ▶ $\beta_1 < 0$ means that increasing *STR* is associated with lower test scores.
- ▶ When $\beta_1 < 0$ and the district reduces *STR*, the predicted test-score change is positive because $\beta_1 \Delta STR > 0$.

The slope connects a change in class size to a change in scores

Move from point A to point B by reducing the student-teacher ratio.

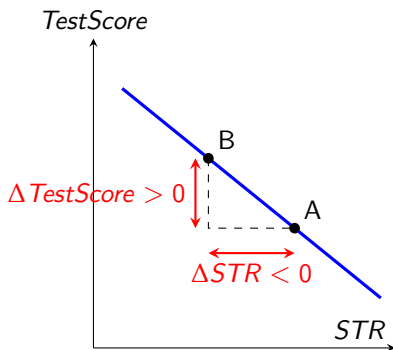
The horizontal movement is

$$\Delta STR < 0.$$

The vertical movement is

$$\Delta TestScore > 0.$$

The ratio of the two changes is the slope β_1 .



Our objective is to use the data to identify this slope.

An exact linear relationship would be simple

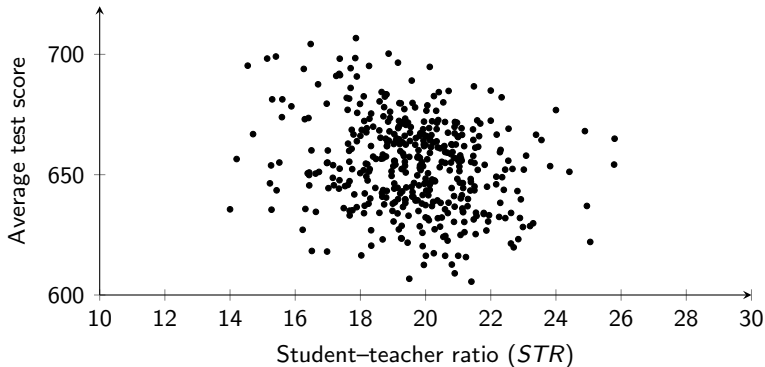
If class size were the only factor determining test scores, every district would satisfy

$$TestScore_i = \beta_0 + \beta_1 STR_i.$$

- ▶ β_0 determines the level of the line.
- ▶ β_1 determines its slope.
- ▶ With $\beta_1 < 0$, districts with higher STR would always have lower predicted test scores.
- ▶ Every observed district would lie exactly on the same straight line.

But this exact relationship is too simple to describe real school districts.

The California observations do not lie on one line



Each dot represents one district. The data do not lie on the same line.

All models are wrong, but some are useful. — George Box

Why do districts with the same *STR* have different scores?

Class size is not the only influence on student achievement.

Test scores may also differ because districts have different:

- ▶ family income and parental education;
- ▶ percentages of English learners;
- ▶ percentages of students eligible for subsidized lunch;
- ▶ teacher experience and quality;
- ▶ school resources and student needs; and
- ▶ other idiosyncratic or random influences.

Our simple line does not explicitly include all of these factors.

The exact linear model does not fit reality because it omits many other determinants of test scores.

Step 4: Represent the other influences with an error

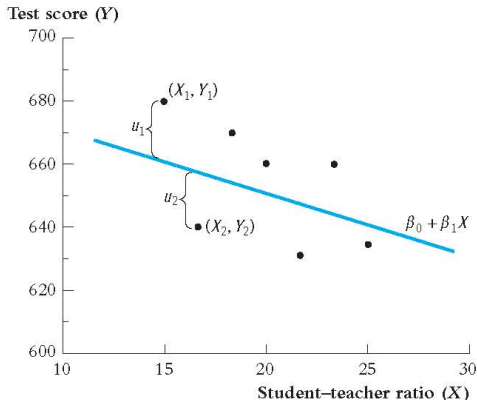
We collect all influences not represented by STR_i in a regression error, denoted u_i .

For a particular district,

$$u_i = TestScore_i - (\beta_0 + \beta_1 STR_i) .$$

- ▶ $\beta_0 + \beta_1 STR_i$ is the score *predicted* by student-teacher ratio, depicted as a linear line.
- ▶ u_i is the (vertical) difference between district i 's *actual score* and that *predicted score* from the line.
- ▶ A positive u_i means the district scores above the line; a negative u_i means it scores below the line.

The regression error is the distance from the population line



For each observed point,

$$u_i = \text{TestScore}_i - (\beta_0 + \beta_1 \text{STR}_i).$$

- ▶ $u_1 > 0$: district 1 lies above the line.
- ▶ $u_2 < 0$: district 2 lies below the line.
- ▶ The errors allow actual observations to differ from the systematic linear relationship.

Step 5: Write the single-variable regression model

Adding the regression error gives the model

$$\boxed{TestScore_i = \beta_0 + \beta_1 STR_i + u_i}, \quad i = 1, \dots, 420.$$

- ▶ Researchers observe $TestScore_i$ and STR_i for every district in the sample.
- ▶ The coefficients β_0 and β_1 are unknown intercept and the slope, called **population coefficients**.
- ▶ Researchers do not directly observe u_i : it summarizes the part of reality omitted from the perfect linear model.
- ▶ The model allows districts with the same STR to have different test scores.

The central econometric task is to estimate β_1

The population slope β_1 is unknown. We use the 420 observed pairs ($TestScore_i, STR_i$) to estimate it.

The estimated slope is denoted

$$\hat{\beta}_1.$$

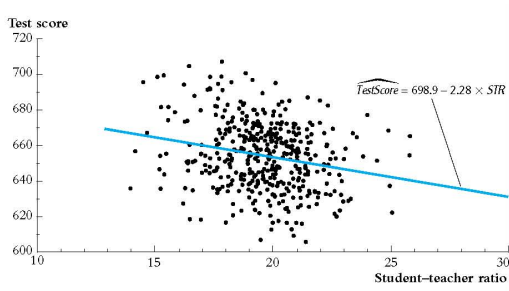
It provides two pieces of information:

- ▶ **Direction:** Is the estimated relationship positive or negative?
- ▶ **Magnitude:** How many test-score points are associated with a one-unit change in STR ?

Regression analysis provides a systematic method for choosing the line and estimating its slope.

The fitted California regression line

The statistical intuition is to choose the line that best fits the observed dots by making their vertical deviations from the line as small as possible.



The estimated line is

$$\widehat{TestScore} = 698.9 - 2.28STR.$$

- ▶ Estimated intercept:
 $\hat{\beta}_0 = 698.9.$
- ▶ Estimated slope:
 $\hat{\beta}_1 = -2.28.$

Population coefficients are fixed, but estimates vary across samples

The population regression line is determined by two parameters:

$$\beta_0 \quad \text{and} \quad \beta_1.$$

These are the **true, but unknown** values that describe the relationship in the population. They are fixed “law of nature” and do not depend on which sample of California districts happen to appear in our dataset.

We use the sample data to estimate these unknown parameters:

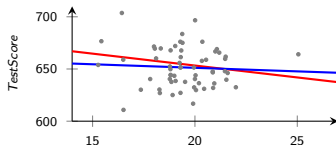
$$\hat{\beta}_0 \quad \text{and} \quad \hat{\beta}_1.$$

These are **estimates** computed from our sample. If we collected a different sample of California school districts, we would generally obtain different estimates—and therefore a different fitted regression line.

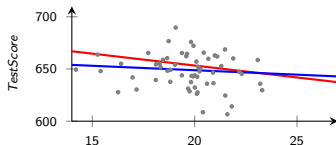
Thus, $\hat{\beta}_1$ is our best sample-based guess of β_1 , but it need not equal β_1 exactly.

Different samples produce different fitted regression lines

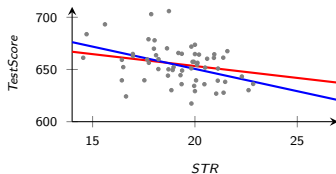
Sample 1: $\hat{\beta}_1 = -0.68$



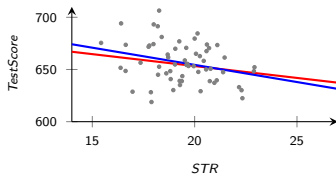
Sample 2: $\hat{\beta}_1 = -0.85$



Sample 3: $\hat{\beta}_1 = -4.27$



Sample 4: $\hat{\beta}_1 = -3.29$



Red: fixed population line, $\beta_1 \approx -2.28$

Blue: fitted sample line, $\hat{\beta}_1$

Each sample contains different school districts of CA and therefore produces a different $\hat{\beta}_1$, even though the population slope β_1 is unchanged.

Econometrics also measures estimation uncertainty

Because $\hat{\beta}_1$ changes from sample to sample, an estimate without a measure of uncertainty is incomplete.

Another central task of econometrics is to quantify:

How uncertain is $\hat{\beta}_1$ as an estimate of the true slope β_1 ?

Later in the course, we will use:

- ▶ **standard errors** to measure the sampling variability of $\hat{\beta}_1$;
- ▶ **confidence intervals** to describe a range of plausible values for β_1 ; and
- ▶ **hypothesis tests** to evaluate claims about β_1 .

Interpret the estimated slope step by step

The estimated slope is

$$\hat{\beta}_1 = -2.28 = \frac{\widehat{\Delta TestScore}}{\Delta STR}.$$

1. The negative sign says that larger student–teacher ratios are associated with lower test scores.
2. If STR increases by one, the fitted test score decreases by 2.28 points.
3. Equivalently, if STR decreases by one, then

$$\widehat{\Delta TestScore} = (-2.28)(-1) = +2.28.$$

4. Thus, districts with one fewer student per teacher have fitted test scores that are 2.28 points higher, on average.

The fitted slope is an association, not yet a causal effect

The regression line summarizes the relationship observed across California districts.

However, districts with lower STR may also have:

- ▶ higher family income;
- ▶ fewer English learners;
- ▶ fewer students eligible for subsidized lunch; or
- ▶ more educational resources.

These factors are part of u_i and may themselves be related to STR_i .

Therefore, $\hat{\beta}_1 = -2.28$ is an estimated **association** between STR and $TestScore$. Later in the course, we will study the assumptions and methods needed to interpret a regression slope as a **causal effect**.